Comparative analysis of the performance of NASA Landsat OLI/TIRS, ESA Sentinel 2 MSI, and Roskosmos Kanopus-V satellites in Klaipėda (Lithuania)

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Remote Sensing, GMS

1. INTRODUCTION

1.1 Study area

Klaipėda ranks as the third largest city in Lithuania. Geographically situated in the western coastal region, the area is generally flat but includes distinctive Lithuanian landscapes: the Curonian Spit with its high sand dunes, and a 90 km coastline featuring sandy beaches and intermittent cliffs. Strategically positioned on the Baltic Sea, Klaipėda serves as a crucial node for domestic and international trade, driven by a prosperous maritime industry and port-related services. It is the site of Lithuania's largest port. The Klaipėda Functional Urban Area extends over 1,326.1 km² and recorded a population of 246,301 at the end of 2024.

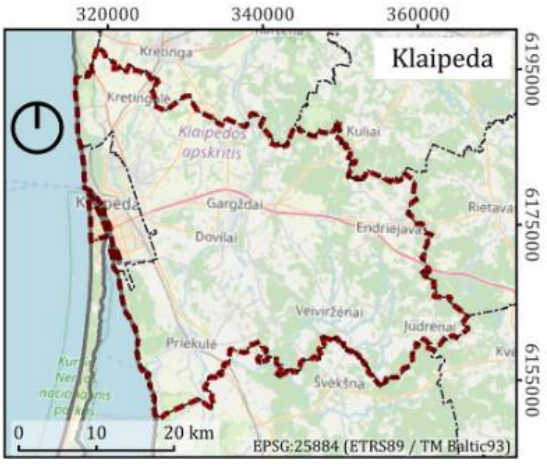


Figure 1. Carte de Klaipéda et ses alentours

1.2 Historical context

Klaipeda is the oldest city in the country. It was founded in 1252 under the name of Memel through the establishment of Memelburg (Memel Castle). After many wars and battles, the territory was occupied by the Swedes and Russians, and then became one of the refuges of the court and the royal family of Prussia. In 1871, Memel was integrated into the German Empire. However, in 1919, after the 1st World War, the Treaty of Versailles stipulated the creation of an autonomous territory, despite the opinion expressed by the population, a large majority of whom considered themselves "German". This led to a revolt in 1923, allowing the territory to be attached to the Republic of Lithuania.



Figure 2. Memelburg

After World War II, when the territory was ceded to Nazi Germany, Klaipeda was part of the Lithuanian SSR of the USSR. Due to its strategic geography on the coastal zone, the region has experienced strong militarization as well as significant socio-economic changes through agriculture and industry.

The fall of the USSR in 1991 is considered a shock in which the country, integrated into the European Union in 2004, experienced a demographic decline, accompanied by a transition to a market economy in the face of unemployment reaching record figures.



Figure 3. The demolition of the fire department HQ in 1981 was the last of the historical building demolitions in Klaipėda. On the left a Soviet propaganda poster declares: Our work is for thee, oh Fatherland. ©Albinas Stubra.

1.3 Territorial issues

Today, the city is urbanized and metropolised thanks to its major asset that is its port, accompanied by the global process of coastal development, but it can also count on tourism, given its unique history. This duality, between a globalised port economy, the driving force behind coastal development and urban concentration, and the enhancement of a historical and natural heritage for tourism development, draws a territory with complex and often contradictory spatial dynamics. On the one hand, the port generates land pressure, intense logistics flows and artificialization of the coastline; on the other hand, tourism and the quality of the living environment require the preservation of landscapes, coastal ecosystems and built heritage. This constitutive tension places development actors in front of a major challenge: reconciling economic competitiveness, residential and tourist attractiveness, and environmental resilience in a limited and coveted coastal area. To meet this challenge, a fine, up-to-date and spatialized understanding of the processes at work is essential. It is precisely here that remote sensing and spatial analysis come into their own. Satellite images, through their ability to provide synoptic, multi-temporal and multispectral data, offer a privileged observatory to decipher the spatial signatures of these dynamics. They make it possible to quantify urban sprawl, to monitor the evolution of land-sea interfaces, to measure port activity, or to assess the state of vegetation and recreational areas.



Figure 4. Curonian Spit

Thus, the transition from the description of the issues to their operationalization for public decision-making requires a methodological leap. This leads us to formulate the central problem of this work.

1.4 Problem

> ***How can a combination of satellite vectors generate concrete geographical information and territorial knowledge essential for the planning and development challenges of Klaipéda?***

To answer this question, we will first carry out a detection, recognition and identification of urban objects and structures, then in a second part, we will make a temporal analysis of recent changes in Klaipėda

2. STATE OF THE ART

To carry out this work, we used 2 sets of data which are satellite images and Open Source data from OverPassTurbo, Copernicus and Lithuanian databases.

2.1 Satellite images

The table below shows the different satellites used and their advantages and disadvantages for our analysis.

Table 1. Comparison of the Landsat 8/9, Sentinel-2A/2B and Kanopus-V satellites

	Landsat 8/9	Sentinel 2A-2B	Kanopus-V
Organisation	NASA/USGS (USA)	ESA (European Union)	Roscosmos (Russia)
Launch Date	L8 : February 2013 L9 : September 2021	S2A : June 2015 S2B : March 2017	July 2012
Spatial Resolution	30 m (Optical MS) 15 m (Pan. Band 8) 100 m (Thermal)	10 m (B2, B3, B4, B8) 20 m (B5, B6, B7, B8A, B11, B12) 60 m (B1, B9, B10)	2.1 m (Panchromatic) 10.5 m (Multispectral)
Revisit	16 days (per satellite) 8 days (L8+L9 combined)	5 days (at equator) 2-3 days at mid-latitudes (2 satellites)	3-5 days (theoretical)
Spectral Bands	11 bands : • 4 VIS + 1 NIR (30m) • 2 SWIR (30m) • 1 Pan (15m) • 2 TIR (10.6-12.5 µm)	13 bands : • 4 VIS at 10m • 6 Red Edge/NIR/SWIR at 20m • 3 atmospheric bands at 60m • No thermal band	5 bands : • 1 Pan (2.1m) • 4 MS (Blue, Green, Red, NIR at 10.5m)
Advantages	• Thermal bands for urban heat island analysis • SWIR bands for urban material discrimination	• Better spatial resolution (10m) • High revisit frequency • "Red Edge" bands for vegetation health	• High spatial resolution (2.1m) • Good for individual building detection • NIR band useful for NDVI

	Landsat 8/9	Sentinel 2A-2B	Kanopus-V
Disadvantages	• Coarse resolution (30m) for urban details • Low frequency • Frequent cloud cover in Baltic region	• No thermal band • Uneven resolution between bands (10m/20m/60m)	• Costly/restricted access • Limited spectral bands • No SWIR or thermal

By manipulating the spectral bands of each satellite, thanks to the "Image Calculator" function of the TerrSet software, we were able to calculate indices. These are images calculated from satellite images that highlight a specific phenomenon that is present, while attenuating other factors that alter the effects on the image. For example, a vegetation index will present healthy vegetation in a light color in the index image. Below, we have listed the indices used before presenting them in the form of a table.

- **NDVI (Normalized Difference Vegetation** Index)
- **NDWI (Normalized Difference Water Index)**
- **MNDWI (Modified Normalized Difference Water Index)**
- **NDBI (Normalized Difference Built-up Index)**
- **BSI (Bare Soil Index)**

Table 2. Spectral indices used in the study

	NDVI	NDBI	NDWI	MNDWI	BSI
Formula	(NIR - Red) / (NIR + Red)	(SWIR1 - NIR) / (SWIR1 + NIR)	(Green - NIR) / (Green + NIR)	(Green - SWIR1) / (Green + SWIR1)	((SWIR1 + Red) - (NIR + Blue)) / ((SWIR1 + Red) + (NIR + Blue))
Purpose	The most classic index for assessing vegetation health, density, and biomass	To map urban and built-up areas	To delineate and monitor open water bodies	An improved water index that better separates water from built-up land by replacing the NIR band with a SWIR band	Designed to enhance the visibility of bare soil and mineral surfaces
Advantages	Easy to calculate and interpret. Values range from –1 to 1 where high value (~0.6 to 0.9) indicate dense, healthy vegetation	Straightforward Calculation : Easy to compute with common satellite bands	Excellent at suppressing vegetation and soil signals, making water stand out clearly	Superior to NDWI : Much more effective at excluding urban and built-up areas, resulting in a cleaner extraction of water bodies	Effective for Soil Mapping: Helps in mapping agricultural fields without cover arid regions, and exposed earth
Disadvantages	In very dense forests or crops, the index can max out and fail to distinguish further growth	Confusion with Bare Soil : Both built-up areas and bare soil have high SWIR	Sensitive to Built-up Areas : Can mistakenly highlight urban surfaces which have a similar spectral response, leading to commission errors	Requires a SWIR band, which is available on most modern satellites (Landsat, Sentinel-2) but not on all	Vulnerable to Confusion : Can be confused with built-up areas (similar spectral response) and is influenced by soil moisture

Finally, we transformed the raw satellite data into usable geographic information through maps, statistics and predictions with the help of the tools for transformation and improvement, classification, forecasting, detection and modeling of changes offered by TerrSet software. Here are the tools used for this analysis :

Table 3. Image enhancement and transformation methods

Module	Category	Definition	Advantages	Disadvantages
MNF	Preprocessing	Transformation that orders components by signal-to-noise ratio, segregating noise in the data	• Better for noisy datasets • Preserves meaningful information in first components	• More computationally intensive than PCA • Still a mathematical abstraction
PCA	Preprocessing	Statistical technique that transforms correlated multispectral bands into uncorrelated components, concentrating variance in the first few components	• Reduces data redundancy • Useful for change detection • Reduces noise	• May lose specific spectral information • Difficult to interpret components physically
RGB	Visualization	Overlay of 3 bands into the Red, Green, and Blue channels to create a color image	• Flexible band combinations for different features • Powerful for photo-interpretation and ground truthing	• Limited to 3 bands at a time • Subjective color interpretation

Table 4. Classification methods

Module	Category	Definition	Advantages	Disadvantages
DECISIONFOREST	Supervised Classification	Modified version of the free-standing DecisionForest module. A machine learning method combining many decision trees for a final vote	• Very high accuracy, often the top performer • Resistant to overfitting • Provides variable (band) importance	Can be slow on very large datasets
ISOCLUST	Unsupervised Classification	Unsupervised classification that uses iterative self-organizing clustering to identify natural groupings in spectral data	• No training data required • Identifie natural spectral classes	• Complex parameterization • Slower than others Unsupervised Classification • Number of classes must be predefined
ISODATA	Unsupervised Classification	Advanced unsupervised clustering that iteratively splits, merges, and eliminates clusters based on statistical parameters	• Better for urban objects • More sophisticated than ISOCLUSTER • Can split and merge classes automatically	• More parameters to set • Results can vary with parameter choices
KMEANS	Unsupervised Classification	Partitions data into K clusters by minimizing within-cluster distance	• Simple algorithm • Effective if classes are compact and spherical	• Requires predetermined number of clusters • Sensitive to initial cluster centers
MLP	Supervised Classification	Supervised neural network classifier or regression with hidden layers that learns non-linear decision boundaries	• Good performance on noisy data • High classification accuracy	• Requires parameter optimization • Long training time
SOM	Supervised Classification	Neural network approach that creates ordered mapping of high-dimensional data onto lower-dimensional space	• Preserves topological relationships • Can reveal subtle patterns • No assumption about data distribution	• No assumption about data distribution • Long training time than DECISIONFOREST

Table 5. Change detection, modelling and projection methods

Module	Category	Definition	Advantages	Disadvantages
CA_MARKOV	Modeling and Projection	Cellular Automata via Markov Model. Simulates land use/cover change by integrating proximity factors, constraints, and transition potentials.	• Integrates spatial drivers (slope, roads, zoning) • Produces change probability maps. • Good for land-use planning	• Requires two historical maps • Complex to calibrate. • Extrapolates past trends.
IMAGEDIFF	Change Detection	Subtracts pixel values of two images to measure change	• Simple and direct. • Very effective for abrupt changes • Visually intuitive	• Highly sensitive to acquisition differences • Requires rigorous radiometric correction
LAND CHANGE MODELER (LCM)	Modeling & Projection	Toolkit for analyzing, modeling, and predicting land cover change with multiple algorithms	• Integrated workflow • Multiple validation tools • Combines multiple approaches	• Requires quality multi-temporal data • Results depend on driver variable selection

2.2 Open source data

- Open Street Map
- CorineLandCover (in Copernicus)
- Lithuanian databases

Then, we retrieved Open Source data from Open Street Map, CorineLandCover and the Lithuanian database. This allowed us to obtain reference data since it is mainly verified, accurate and recognized official data. They then serve as an example for our analyses and processes carried out on the TerrSet software, allowing us to sometimes observe limits and flaws in certain data. Finally, we were able to use this data to compare it with our results in order to estimate validation percentages on the consistency of the results. The table below presents the advantages and disadvantages of this OpenSource data for our work.

Table 6. Advantages and disadvantages of the open source data

	Open Street Map	Corine Land Cover	Lithuanian databases
Advantages	• Accès libre et gratuit • Mises à jour fréquentes par contributeurs locaux • Informations attributaires riches • Caractéristiques contributives	• Accès libre et gratuit • Données officiellement vérifiées et validées • Nomenclature harmonisée	• Source autoritaire • Données officiellement vérifiées et validées • Haute précision positionnelle • Informations cadastrales complètes • Dates de construction et permis • Hauteurs et surfaces de bâtiments
Disadvantages	• Pas de validation officielle • Dépend de l'effort bénévole • Dates de construction souvent manquantes	• Résolution spatiale grossière • Détails urbains très limités • Pas d'information inter-période • Changements récents (post-2018) non disponibles • Pas de distinction types de bâtiments	• Détails limités sur petites structures urbaines

3. METHODOLOGY

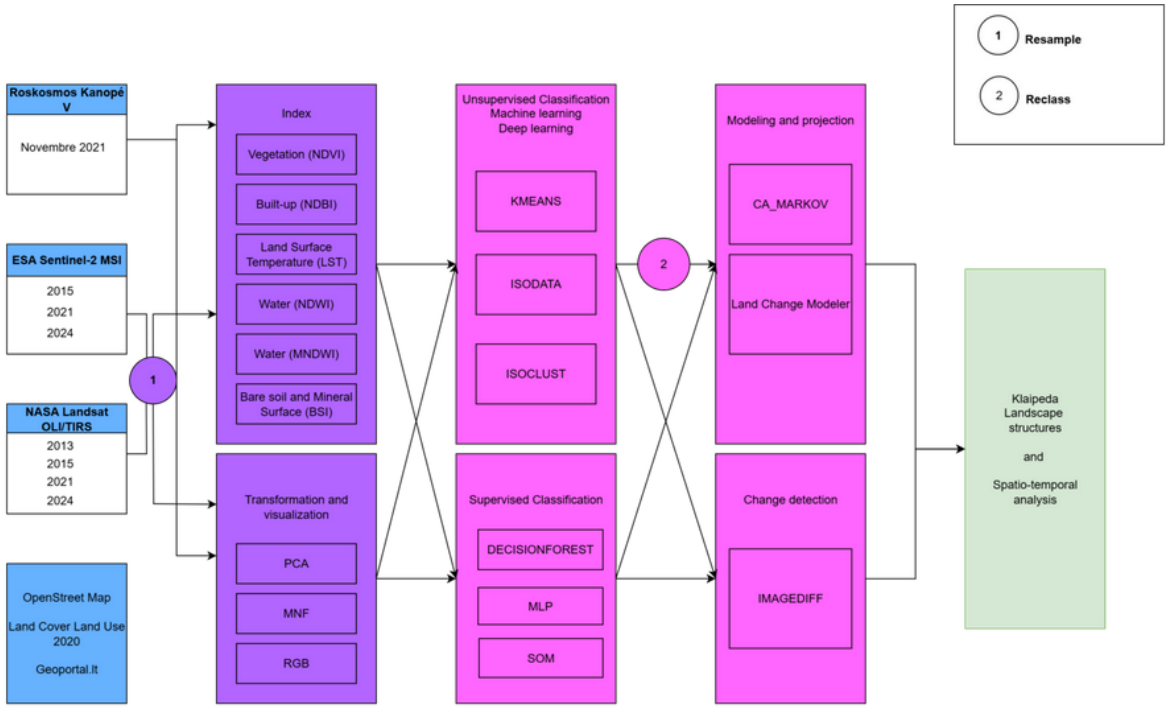


Figure 5. Methodological workflow

4. ANALYSIS OF RESULTS

4.1 Detection, recognition and identification of urban objects and structures

In this part, we focused on the urban objects of Klaipeda. First, it was necessary to carry out a typology of urban object types, structures and morphologies for recognising and mapping urban land use types and functions. Finally, we made comparisons between these results.

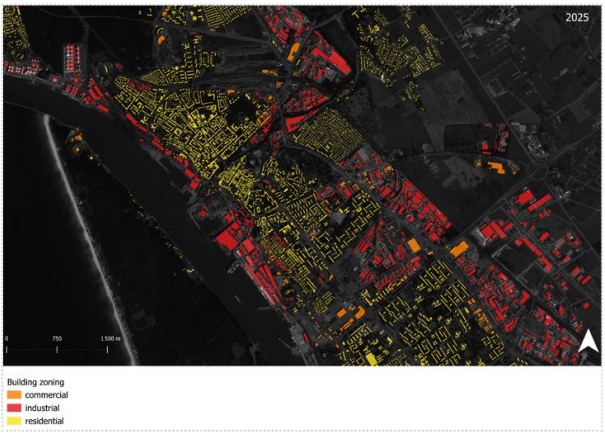


Figure 5 : Urban structure of Klaipėda

Figure 6. Urban structure of Klaipėda

The map shows an urban organization typical of a Baltic port city. The visualization reveals a residential building

clearly dominating the space, particularly in the heart of the agglomeration. This residential building is formed according to a dense orthogonal architecture, particularly in the centre, probably testifying to the heritage of the city's history. The historic centre is easily identifiable in particular by its morphology characterised by an irregular network of winding and narrow streets, although the city was heavily destroyed during the Second World War, its road network has been reconstructed. It is bounded by the port and the river. To the south-east of the latter, there is a residential area made up of collective complexes, identifiable by the size and atypical shape of the collective buildings. In peri-urban areas, we find spaces made up of individual buildings (in the shape of small dots here on this scale), especially in the north of the city. Industrial areas are mainly polarized on the outskirts to the east, but also along the port, encircling residential areas. It should be noted that this industrial building does not form a single large industrial zone. It is identifiable, for the most part of the buildings, by their size, they are buildings with a large footprint designed to house storage lines, large machines or production, but also by the shape and material of their roofs.

Commercial buildings, which are more dispersed, are mostly located along the main roads, often near residential buildings, which suggests that access to local services is easier, thus reducing travel for services and everyday purchases. We therefore have a development policy that favours contact between place of residence, work and consumption while maintaining a certain isolation from industrial activities.

However, this proximity between the residential and industrial sectors can raise problems of noise, traffic or atmospheric pollution, or otherwise problems of incompatible uses. The city must therefore pursue development policies that reduce these problems of use, but it must also find a way to reconcile its port economic activities with its urban quality of life in order to ensure its preservation. This could potentially push Klaipeda to face the challenge of the ecological transition, which will involve decarbonisation objectives.

4.2 Temporal analysis of recent changes in Klaipėda

In this part, we have Create a time series of Sentinel 2 MSI and Landsat 8/9 OLI/TIRS data from 2013 to 2025 at the same period to observe the evolution of the territory, identify the types of changes. Finally, we made comparisons between these results.

4.2.1 Landsat 8/9 OLI/TIRS



Figure 7. Vue satellite de Klaipéda (2013)

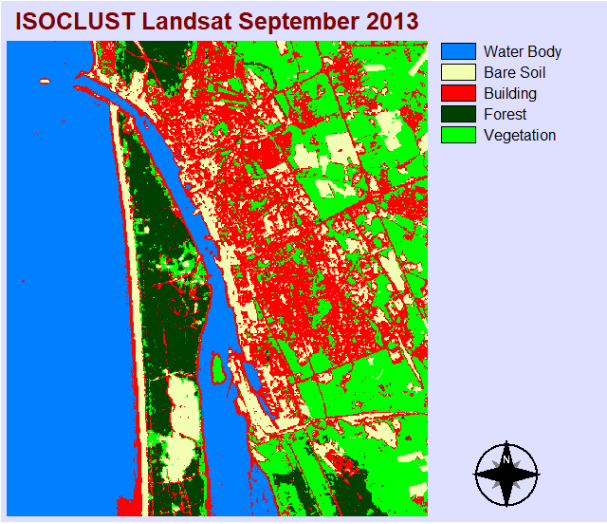
We have chosen to keep 4 dates : 2013 ; 2015 ; 2021 ; 2024. These images were taken during the same period (between September and November). After having carried out several classification methods for each date, we made projections for the future but also projections for 2024-2025 and 2035-2055 in order to observe (or not) similarities between the predictions and reality.

4.2.2 Landsat 2013

Unsupervised classification

This ISOCLUST was made from 7 spectral bands of Landsat (B2-B8), 70 iterations and 5 desired output clusters.

The result shows more or less homogeneous classes with little noise and fairly well differentiated.



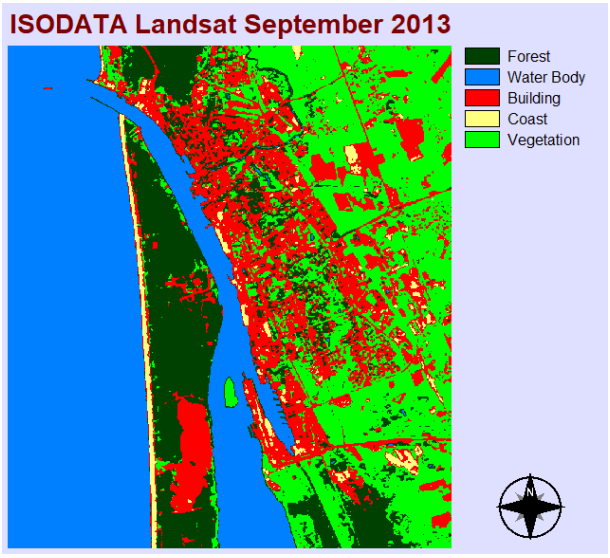
Area on file: ISOCLUST 5 classes

Category	Hectares	Legend
1	4873,027500	Water Body
2	1794,105000	Bare Soil
3	3571,807500	Building
4	1327,567500	Forest
5	3158,842500	Vegetation

Figure 8. ISOCLUST classification of the Landsat scene (September 2013) and area per class.

This ISODATA was made from 7 spectral bands of Landsat (B2-B8), 70 iterations and an initial and output number of 5 clusters.

The result demonstrates an acceptable overall spatial coherence with more or less homogeneous classes



Area on file: ISODATA 5 classes

Category	Hectares	Legend
1	2362,905000	
2	5304,622500	
3	2887,627500	
4	308,385000	
5	3861,810000	

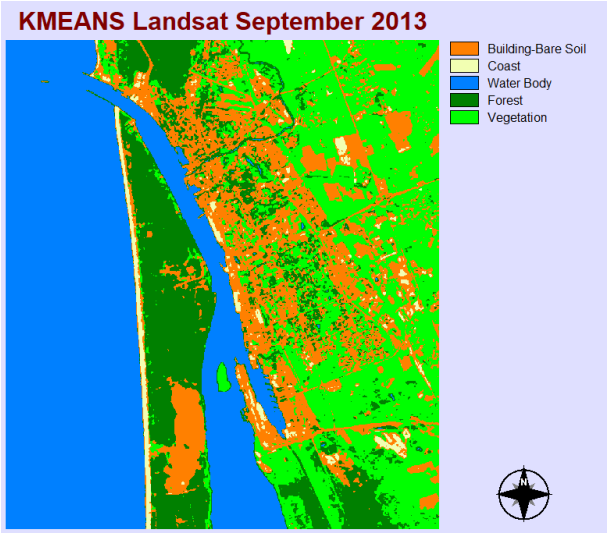
Figure 9. ISODATA classification of the Landsat scene (September 2013) and area per class.

The Forest is well identified, as is the ISOCLUST.

The spatial continuity is remarkable. It should also be noted that there are small urban woodlands. The Water Body is also similar to the ISOCLUST, except for the harbour which has been well identified, as well as the canal to the north of the town and some inland waterholes. As for the Building, it has significant confusion with the Bare Soil and part of the Vegetation but still identified urban and industrial areas. The Coast, a new cluster, represents the coastal sandy area, as well as white roofs and some of the industrial buildings. Finally, the Vegetation has a good identification despite nuances. This cluster is similar to that of ISOCLUST, however, some parts of the Bare Soil have been confused with the Vegetation.

This KMEANS was made from 7 spectral bands of Landsat (B2-B8), 70 iterations and an initial and output number of 5 clusters.

The result demonstrates a more spatially homogeneous classification.



Area on file: KMEANS 5 classes

Category	Hectares
1	2889,945000
2	301,657500
3	5304,442500
4	2365,807500
5	3863,497500

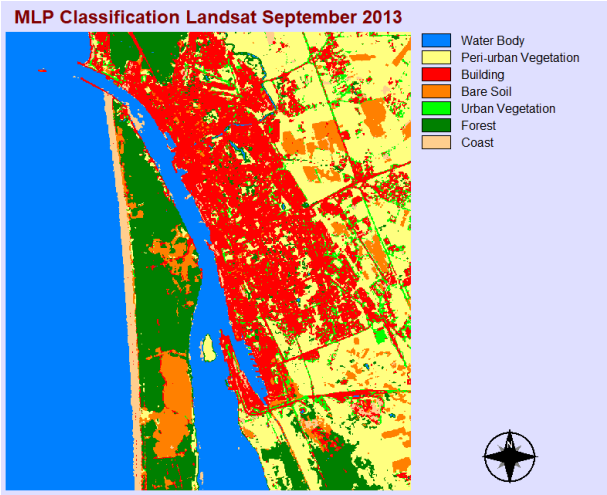
Figure 10. KMEANS classification of the Landsat scene (September 2013) and area per class.

The Building and the Bare Soil have been grouped together, as in ISODATA above. The distinction was once again not made. The Coast has a similarity to ISODATA with the same qualities and flaws. This goes hand in hand with the Water Body, the Forest and the Vegetation.

Supervised classification

This MLP was made from 7 spectral bands of Landsat (B2-B8) and 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI).

The result demonstrates, through a sophisticated and multivariate approach, a methodological improvement compared to previous unsupervised classifications



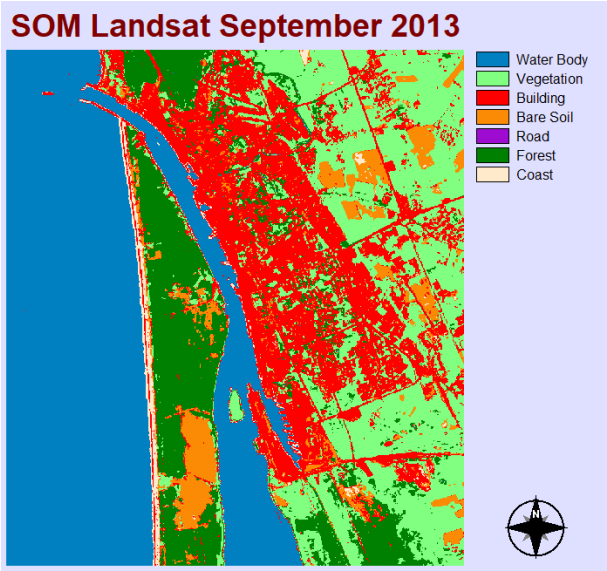
Area on file: MLP Classification Bandes Spectrales et Indices

Category	Hectares	Legend
1	5186,835000	
2	2961,180000	
3	3031,762500	
4	880,695000	
5	516,172500	
6	1698,817500	
7	449,887500	

Figure 11. MLP classification of the Landsat scene (September 2013) and area per class.

This SOM was made from 7 spectral bands of Landsat (B2-B8), 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI) and a MAKESIG with 7 clusters.

The result is an overall satisfactory classification with a good performance for the natural classes (Water Body, Vegetation and Forest. On the other hand, the performance is not as good for the rest.



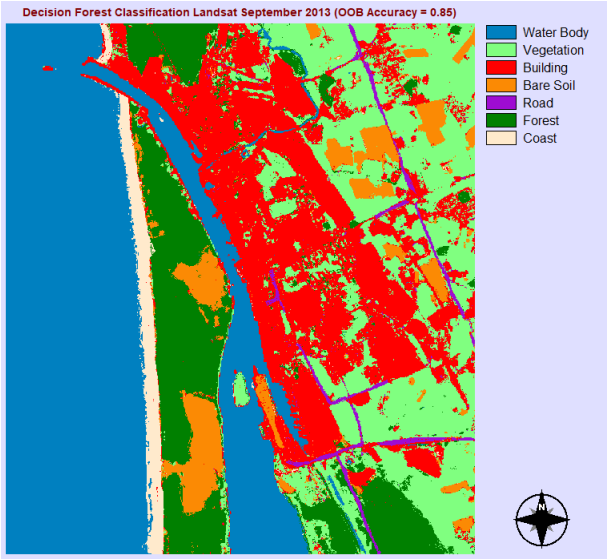
Area on file: SOM Bandes Spectrales et Indices

Category	Hectares	Legend
1	5152,095000	
2	3354,007500	
3	3241,890000	
4	839,902500	
6	1921,612500	
7	215,842500	

Figure 12. SOM classification of the Landsat scene (September 2013) and area per class.

This DECISIONFOREST was made from 7 spectral bands of Landsat (B2-B8), 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI) and a MAKESIG with 7 clusters.

The classification by Decision Forest has an OOB (Out-Of-Bag) accuracy of 0.85 (85%), which is an excellent performance indicator



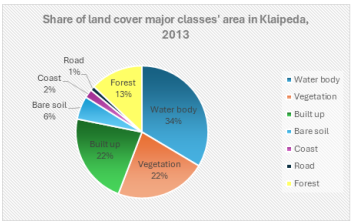


Figure 13. DECISIONFOREST classification of the Landsat scene (September 2013) and area per class.

The Water Body has a better classification than the SOM since we can distinguish 2 channels located in the North and South. The Vegetation mixes with the Building, especially in the city centre. The Bare Soil is less fragmented. The Road has been detected where major roads can be distinguished. The Forest is less pronounced here in the city center. It is still confused with the Vegetation. Finally, the Coast is better represented, the distinction with the Building is well made.

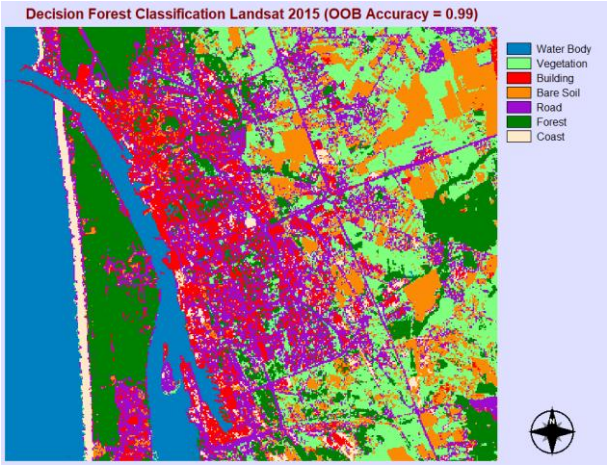
These 3 supervised classifications show us more or less different results. They create a complete processing chain. Only the DECISIONFOREST has succeeded in distinguishing the Road, in particular due to the fact that the spatial context is very little taken into account. In SOM, semantic flexibility is complex. As for the MLP, it differentiates the type of vegetation

4.2.3 Landsat 2015

Supervised classification

This DECISIONFOREST was made from 3 spectral bands of Landsat (B2, B3 et B4) and a MAKESIG with 7 clusters.

The result is not very suitable, without any spatial coherence.



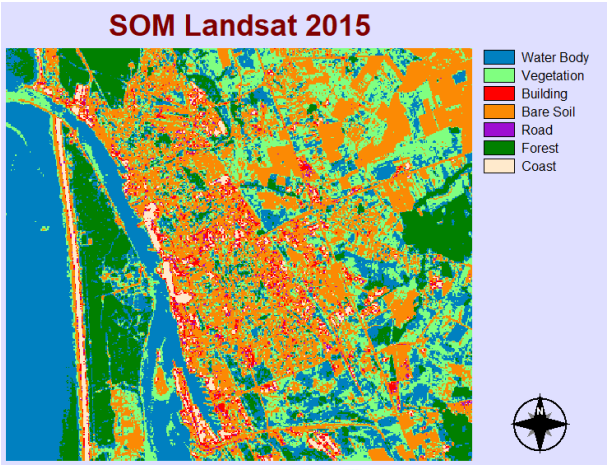
Area on file: FOREST__df

Category	Hectares
1	2583,607500
2	2637,697500
3	1503,945000
4	1771,425000
5	2909,700000
6	2353,230000
7	531,315000

Figure 14. DECISIONFOREST classification of the Landsat scene (2015) and area per class.

This SOM was made from 3 spectral bands of Landsat (B2, B3 et B4) and a MAKESIG with 7 clusters.

The result is worse than the DECISIONFOREST, with spatially inconsistent classifications.



Area on file: SOM_2015_RGB_Supervised

Category	Hectares	Legend
1	4224,217500	
2	2907,877500	
3	385,830000	
4	4359,240000	
5	155,475000	
6	1821,465000	
7	436,815000	

Figure 15. SOM classification of the Landsat scene (2015) and area per class.

In this SOM, only the Forest is more or less well represented. The rest is misclassified. The Road has not even been identified. The result is once again bad.

4.2.4 Landsat 2021

Supervised classification

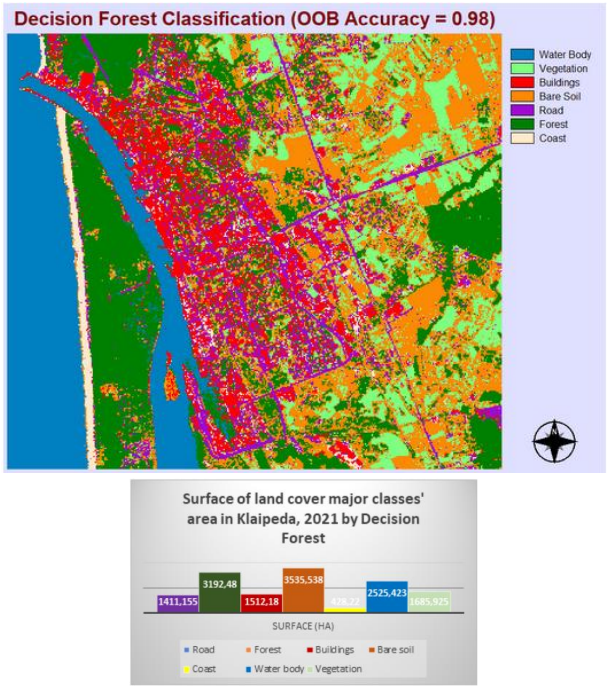


Figure 16. DECISIONFOREST classification of the Landsat scene (2021, OOB accuracy = 0.98) and area per class.

Once again, the "Decision Forest" method gives very good results, with excellent performance, illustrating the remarkable reliability of the Machine Learning algorithm. These results highlight the occupation of the territory of Klaipėda, a Baltic port city, with a clear distinction between the major spatial areas of occupation: the body of water in the west, the urban fabric at the heart of the map and the green spaces that dominate the peri-urban and forest areas. Once again, bare soils probably reveal agricultural areas or simply areas in urban transition, identifiable in particular thanks to infrared data from Landsat satellites.

However, although the OOB accuracy is remarkable, with a value of 0.98, the visualization shows a "salt and pepper" effect, especially in anthropogenic spaces where the pixels of the different sets intermingle in a fragmented way. This is most likely due to the low resolution of the Landsat satellite bands. At this resolution, a pixel can contain different elements of spatial occupation (vegetation, roof, road, etc.), thus generating mixed spectral signatures that reveal this fragmentation in heterogeneous environments, limiting the fine interpretation of the spatial structure of the territory, particularly urbanized spaces. But this method remains, despite everything, interesting for the analysis of global spatial occupation, as can be seen with forest areas, which are remarkably accurate.

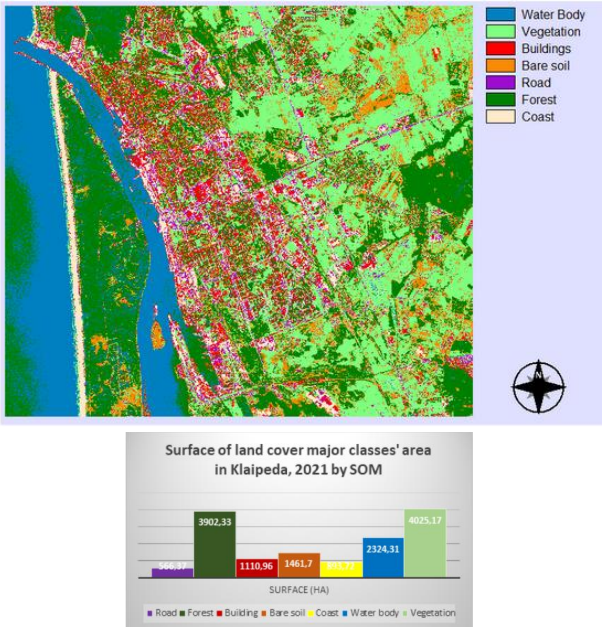


Figure 17. SOM classification of the Landsat scene (2021) and area per class.

The classification by Self-Organizing Maps applied to the RGB result of the Landsat bands offers broadly the same results as the previous method. Indeed, the visualization clearly makes it possible to identify and distinguish the plans present in the region, in particular that of the Curonian lagoon, the vegetation, the urban spaces, marked by a significant presence of buildings in industrial areas, the city center but also residential areas. But above all, it makes it possible to identify forest areas. The distinction between low vegetation and forest is very interesting because it makes it possible to differentiate vegetated areas according to their nature or density. The road network, although very fragmented, is easily identifiable, as well as the bare ground areas, mainly occupying the eastern and southern zones.

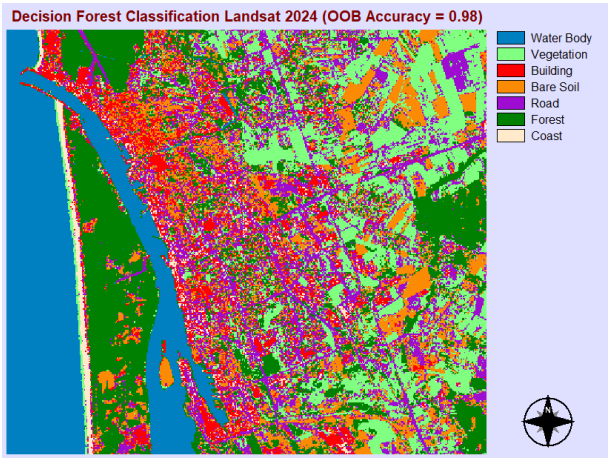
Nevertheless, despite the fact that this classification is more or less valid, we find, here too, the "salt and pepper" effect, which remains very pronounced, more precisely in dense urban spaces. The result is a significant fragmentation of classes, giving rise to spectral confusions, particularly between roads and buildings, but also between bare soils, the "coast" and vegetation in urban spaces.

Although it is interesting to provide the natural structure of spectral data, the SOM method does not offer a robust enough quantitative validation like the OOB precision in Decision Forest. As can be seen on the map, this classification by Self-Organizing Maps reveals several errors related to the spectral signatures of certain elements, in particular the presence of elements of the "Forest" class in an area in the Baltic Sea to the west of the city, but also the confusion between the built environment and the "Coast".

In any case, regardless of the method used, post-processing of the data is still necessary to reduce the fragmentation related to the resolution of the Landsat

OLI/TIRS satellite bands, thus offering much more usable graphical results.

Modeling and projection

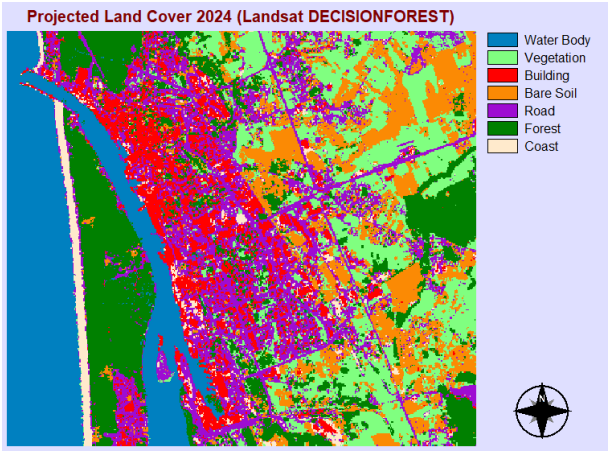


Area on file: Forest_Landsat_2024_df

Category	Hectares
1	2581,582500
2	2638,192500
3	1198,395000
4	2049,345000
5	2536,605000
6	2965,815000
7	320,985000

Figure 18. DECISIONFOREST classification of the Landsat scene (2024, OOB accuracy = 0.98) and area per class.

This is a DECISIONFOREST made from 3 spectral bands of Landsat (B2, B3 and B4) in 2024. We then compared this image to the projection CA_MARKOV made using the 2015 and 2021 DECISIONFOREST to find out whether or not the result of the CA_MARKOV is consistent with the 2024 DECISIONFOREST.



Area on file: CA_Markov_L8_2015_2021_Prevision3ans5x5

Category	Hectares
1	2553,007500
2	2519,122500
3	1417,545000
4	2003,625000
5	2680,425000
6	2631,105000
7	486,090000

Figure 19. CA-Markov projection of land cover for 2024 (from the 2015 and 2021 classifications) and area per class.

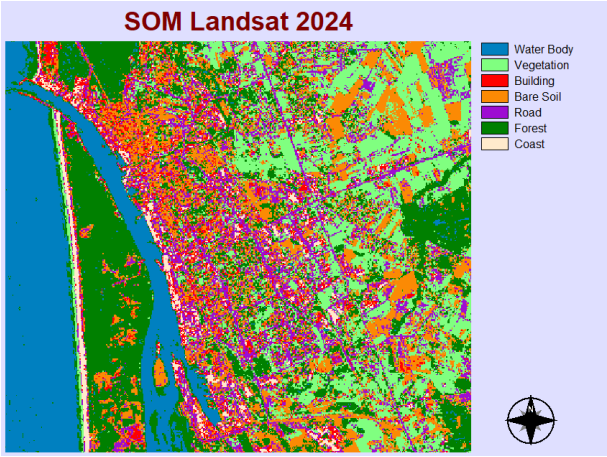
Comparing the acreage results, the prediction is roughly good for the Water Body, the Vegetation, the Bare Soil, and the Road. The results are more or less equivalent. On the other hand, the Building is over-represented, unlike the Forest which is under-represented. The Coast cluster has undergone an increase in prediction that should not be overlooked. All in all, the prediction is quite correct, even if some clusters have a different representation. This prediction still needs to be qualified for clusters with equivalent results since in some places, the land cover is different compared to the real DECISIONFOREST, especially for the Vegetation and the Bare Soil.

4.2.5 Landsat 2024

Supervised classification

This SOM was made from 3 spectral bands of Landsat (B2, B3 et B4) and a MAKESIG with 7 clusters.

Le résultat est semi-réaliste, avec des classes plus ou moins plausibles.



Area on file: SOM_RGB_Landsat_2024_Supervised

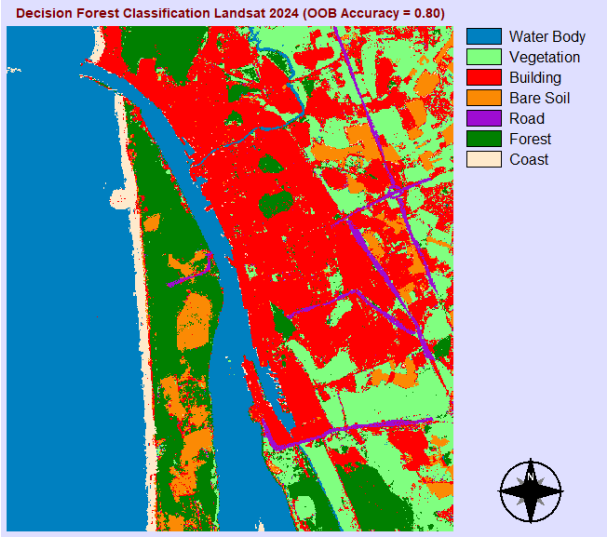
Category	Hectares	Legend
1	2431,867500	
2	2883,172500	
3	838,125000	
4	2524,050000	
5	1902,645000	
6	3298,950000	
7	412,110000	

Figure 20. SOM classification of the Landsat scene (2024) and area per class.

The Water Body has been captured, whether it is the Sea, the lagoon or the land waterholes. The vegetation is moderately realized. Peri-urban vegetation is well characterized, unlike urban vegetation which is not. The Building is not well represented and has been partly confused with the Bare Soil, hence the large surface area of this cluster. The Roas is also overestimated, if you can make out the road network, you can see plots in the city centre and on the outskirts. The Forest is well detected, as for the Coast, the sandy area is represented, but the industrial and port areas are also represented, as are some buildings.

This DECISIONFOREST was made from 7 spectral bands of Landsat (B2-B8), 5 indices (MNDWI, NDBI, BSI, NDVI, NDWI) and a MAKESIG with 7 clusters.

Le résultat présente une bonne robustesse, comme l'indique l'OBB. Les classes sont plus distinctes.



Area on file: DecisionForest Indices_df

Category	Hectares
1	5195,745000
2	2201,062500
3	4020,952500
4	824,962500
5	278,662500
6	1875,240000
7	328,725000

Figure 21. DECISIONFOREST classification of the Landsat scene (2024) and area per class.

The Water Body is still well represented. The Vegetation, with the help of NDVI, is well distinguished from the Bare Soil. There are small areas of the Vegetation in the city center, like for the Forest. The Building has a good spatial coherence both in the centre and on the outskirts. As for the Road, it is better represented. Finally, the Coast is only located on sandy areas, showing the good realism of this cluster.

Modeling and projection

From now on, we wanted to make long-term predictions, through the CA_MARKOV tool, using the DECISIONFOREST of 2013 and 2024 in order to estimate land use for the years 2035 and 2055, in contexts marked by urban sprawl and sea level rise, 2 phenomena that can impact Klaipéda and its territory.

The Markov model application shows a nuanced projection of how land use changes over thirty years, highlighting continuity and transition dynamics. The classifications established in 2013 and 2024 served as the basis for this projection and reveal a landscape with a strong presence of dense natural masses. The results indicate an overall persistence of this fundamental structure, dominated by forests, vegetation cover and water bodies, thus highlighting the resilience of these large complexes to the pressure of urbanization in the short and medium term.

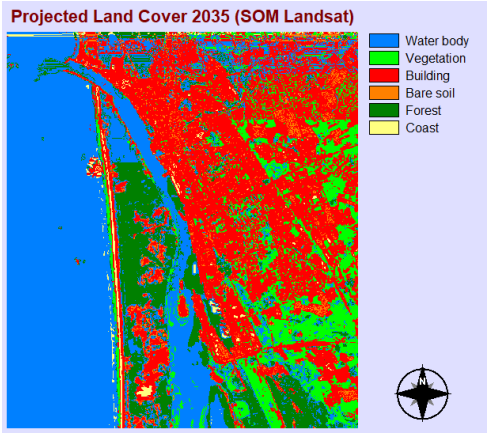


Figure 22. Projected land cover for 2035 (SOM, Landsat).

The 2035 projection presents a first stage in the evolution compared to 2025. It can be seen that the fundamental structure of the region, which are the forest areas, the water bodies and the vegetation cover, remains generally preserved, thus highlighting a certain inertia of these major ensembles. However, the same cannot be said for bare surfaces. They indicate major spatial variations, which may reflect the dynamics of construction sites, temporary abandonment or land clearing. The latter seem to experience localized gains, with probable extensions and/or densifications at the periphery of existing nuclei, which follow the calculated transition probabilities. This projection marks a period of moderate change, in which the trends of territorial change, despite the fact that they have already begun, have not yet dramatically altered the balance of the territory. It represents a crucial step, which validates the orientation of the model before the longer-term projection.

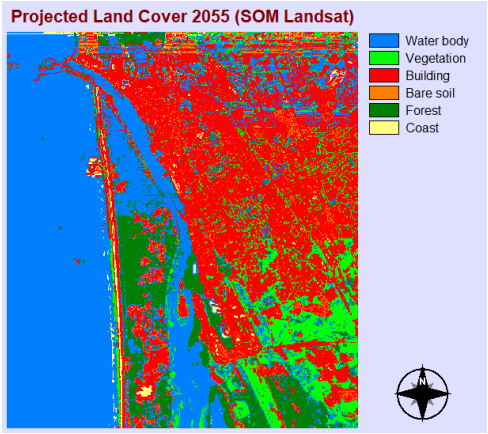


Figure 23. Projected land cover for 2055 (SOM, Landsat).

The projection for 2055 shows an exacerbation and a reinforcement of the dynamics started in the 2035 projection, giving a vision of a region whose land use has undergone significant changes over three decades. The resilience of large natural areas, wetlands, forests and coastlines, consolidates their sustainable disposition, despite the fact that there is localized retreat and fragmentation in their fine distribution, particularly in areas close to areas of urban pressure. Concerning artificialized areas, we observe a more marked, densified and more saturated morphology, particularly towards the east of the territory. Indeed, the urban task tends to

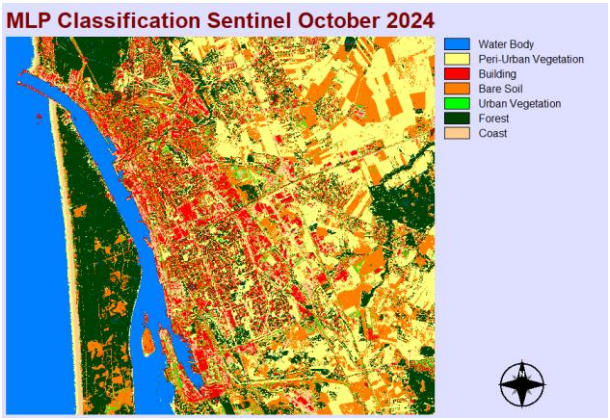
expand towards unbuilt spaces according to the logic of spatial continuity, thus often reducing bare ground spaces. We can also note the decline of low vegetation in favour of the extension of buildings. This projection shows a clear shift in the balance between natural and artificial in some areas of the Klaipeda region, which materializes the cumulative influence of thirty years of transitions and modelled changes, and underlines the influence of the trends present in the projection and future organization of this region.

4.2.6 Sentinel-2 MSI

Supervised classification

Made with Sentinel B2, B3 and B4 tapes.

The map is consistent for a classification with 3 visible bands, but some classes seem to be not very distinct spectrally



Area on file: MLP_2024LULC_Test_812

Category	Hectares	Legend
1	2559,440000	
2	3596,230000	
3	1470,280000	
4	2529,330000	
5	418,430000	
6	2786,400000	
7	924,450000	

Figure 24. Supervised classification of the Sentinel-2 scene (bands B2, B3, B4) and area per class.

The Water Body has a good realism taking into account the sea and the lagoon. A distinction has been made between the Vegetation, however, the Urban Vegetation is very much in the minority. The representation of the Bare Soil is correct. Concerning The Forest, the precision is good, the same for The Coast.

Modeling and projection

The application of the Markov model to the city of Klaipėda shows a nuanced projection of land-use evolution over a thirty-year period, highlighting both continuities and transition dynamics. Classifications produced for 2015 and 2024 using Sentinel-2 satellite data served as the basis for this projection and reveal a landscape characterized by a strong presence of dense natural areas. The results indicate an overall persistence of this fundamental structure, dominated by forests, vegetation cover, and water bodies, thereby emphasizing the resilience of these major land-cover units to urbanization pressure in the short to medium term. Below is an example of a classification produced using the "Decision Forest" algorithm for the year 2024, providing the most accurate and clear representation of land use in the city of Klaipėda, in order to compare the results obtained for future projections.

Based on the data presented on our map, we obtain the following land-use distribution :

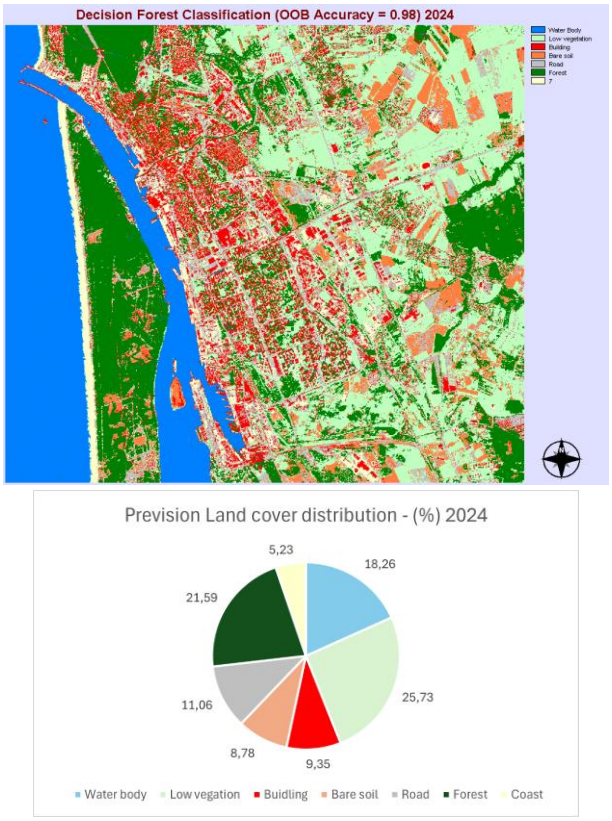


Figure 25. DECISIONFOREST classification (OOB accuracy = 0.98) of 2024 and land cover distribution (%).

Before carrying out various projections for the coming years, we decided to produce a forecast for the year 2024 using our 2015 and 2021 data in order to assess the consistency of these future projections. To enable comparison, we analyzed the differences between the two results, particularly in terms of surface area by classification, since the two models were visually very similar. A CROSSTAB analysis was also performed to compare the number of differing pixels.

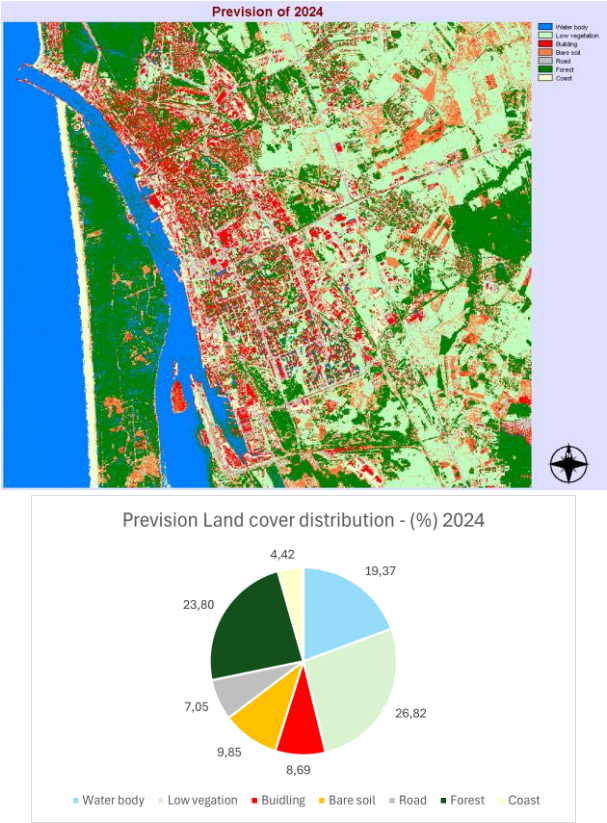


Figure 26. Forecast for 2024 (from the 2015 and 2021 classifications) and land cover distribution (%).

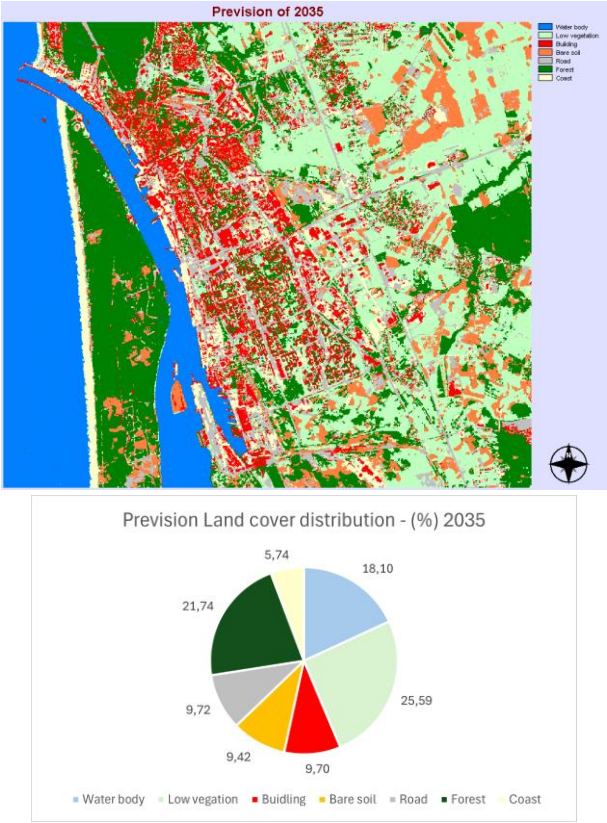


Figure 27. Projected land cover for 2035 and land cover distribution (%).

As previously stated, the projection corresponding to the year 2024 is very close to the observed reality, with a few notable differences, such as the presence of water along the western coast and in the city center, or discrepancies related to bare soil areas. However, a more detailed examination of the changes between these two maps is required.

By Analyzing the different percentages, it becomes apparent that the similarities are not only visual but are also reflected in the land-use percentages. Indeed, the largest difference between the two maps concerns the road class, with a difference of only 4%. This analysis demonstrates that the forecasting system is reliable and provides insight into the level of uncertainty that may be expected in the results.

A projection for 2035 presents an initial stage of evolution compared to 2024. It can be observed that the fundamental structure of the region comprising forested areas, bodies of water, and vegetation cover remains largely preserved, highlighting the inherent inertia of these major components. This forecast is very close to the reality in 2024. Notably, there is a slight increase in buildings, a decrease in bare soils and roads (likely due to inaccuracies in the model), and even an increase in areas occupied by forests.

We can observe this consistency in the results provided by the land use data of the city of Klaipėda:

The distribution of this area remains consistent with our previous models. The differences between the 2024 forecast and that of 2035 can be explained by the fact that the 2024 forecast did not use data from 2024. As mentioned earlier, this forecast is based on an analysis of data from 2015 to 2021, which explains the notable difference between the two models. The data used for the 2035 and 2055 forecasts are from 2015 and 2024, which naturally leads to a similarity with the 2024 results of "DECISIONFOREST.".

For the 2055 forecast, we can observe much greater imprecision in the model. Pixel groups increasingly correspond to a single class and are grouped more coarsely. This can be explained by the decreasing robustness of the model over time: the further ahead we try to make forecasts, the less precise they will be, as the quality of the input data declines due to differences in dates. The model also shows certain inconsistencies, such as the new appearance of forests on waterways or around the port. Even though this model is much less precise, it still allows us to make estimates regarding the city's urbanization and how it is likely to develop.

Here, the land use data for Klaipėda appears more realistic than the visual representation. Indeed, the figures predict a decrease in low vegetation and forest areas in favor of an increase in building coverage, roads, and bare soils. However, we have some doubts regarding the changes in water and coastal areas. For water, as seen previously on our map, there is a lot of "noise" along the waterways, which seems far from a likely reality. Doubts

regarding the coast arise from its initial imprecision, as it is difficult to distinguish from certain buildings.

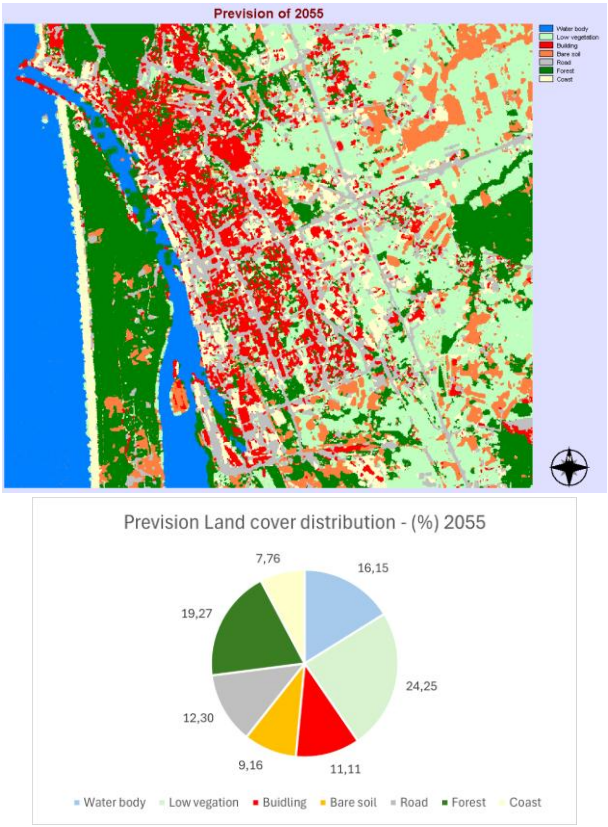


Figure 28. Projected land cover for 2055 and land cover distribution (%).

4.3 Comparison between sensors (SOM classification, 2021)

To analyze the differences between our various satellites, we chose to use the SOM algorithm to perform a supervised classification. We logically selected 2021 for this comparison, as it is the only year for which we have data from Landsat, Sentinel-2, and Kanopus.

The decision to use the Self-Organizing Map (SOM) algorithm was made to ensure all three datasets were analyzed using the same supervised classification method. Since SOM yielded the best results after 'DECISION FOREST,' we opted for the former; 'DECISION FOREST' could not be implemented on the Kanopus data due to the image's pixel count.

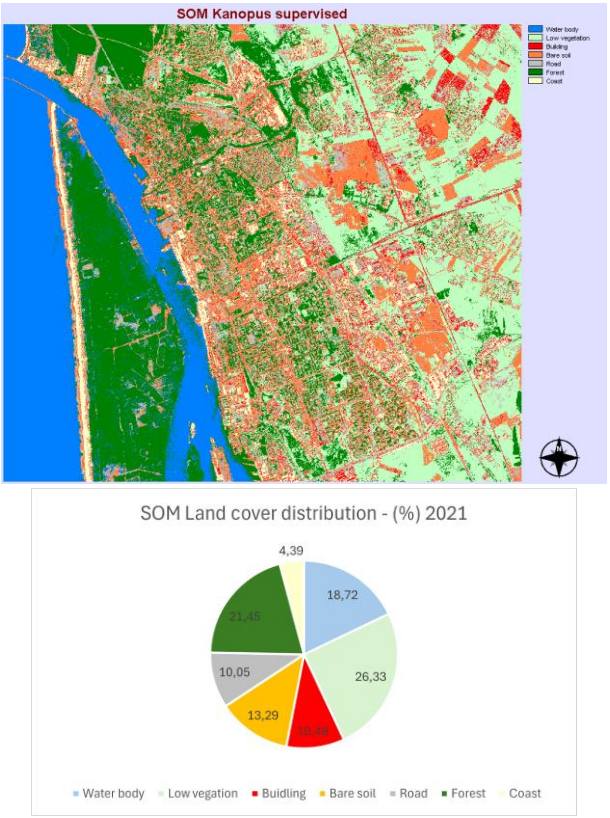


Figure 29. SOM supervised classification of the Kanopus-V image (2021) and land cover distribution (%).

The results obtained using Kanopus spectral bands demonstrate high precision, thanks to the data quality and spatial resolution. However, building detection proved to be inaccurate. On the map, the 'building' class is frequently confused with roads or bare soil, which is expected given their similar spectral signatures.

A more significant issue occurs where bare soil (in the top right of the map) is misclassified as a building. The 'coastline' class faces a similar, albeit less severe, problem. Conversely, we observe high accuracy in forest detection and the delineation between the treeline and low vegetation. Given that the high spatial resolution provides such fine detail, it is also possible that errors were introduced during the model training phase when selecting the class samples.

Regarding land cover, roads occupy approximately 25% of the surface area. This figure is quite close to the data observed in 2024 using the 'Decision Forest' model. Nevertheless, classification errors remain visible in this model, ironically highlighted by the high level of precision it offers.

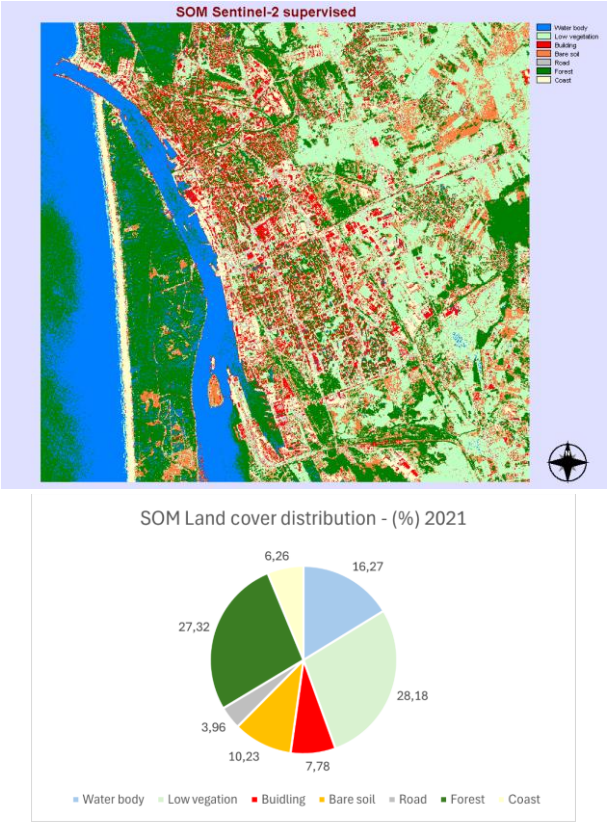


Figure 30. SOM supervised classification of the Sentinel-2 image (2021) and land cover distribution (%).

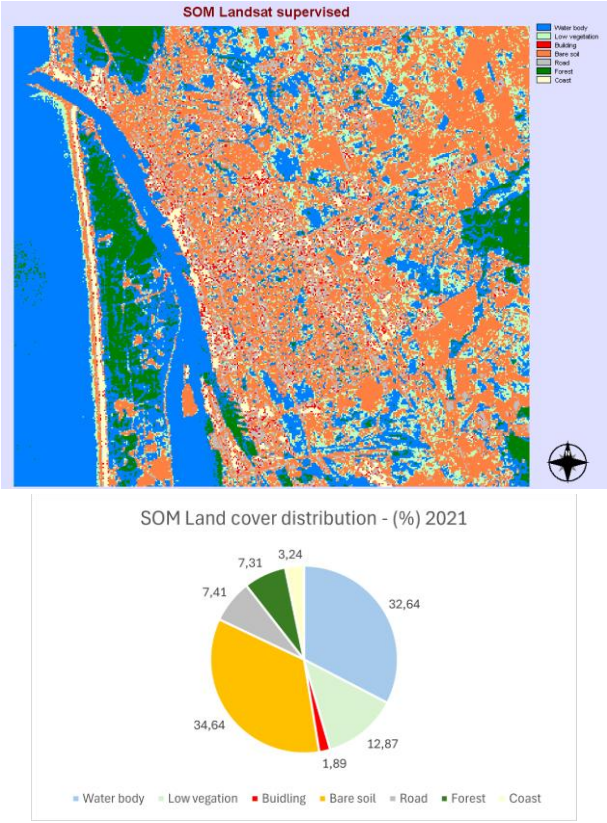


Figure 31. SOM supervised classification of the Landsat image (2021) and land cover distribution (%).

The results obtained with the Sentinel-2 spectral bands are visually very accurate, particularly regarding building detection. However, some issues persist, such as the misclassification of forests within watercourses or the assignment of the 'coastline' class to roads and certains buildings.

Nevertheless, the class distribution remains significantly better than that of the Kanopus classification. A recurring observation across the three SOM runs is that as the spatial resolution increases, these types of classification errors become more frequent. Overall, the results between Kanopus and Sentinel-2 remain highly consistent, with a slight preference for Sentinel-2, as its outputs are already very close to ground truth.

Here, we observe the land cover distribution resulting from the SOM performed with Sentinel-2 spectral bands. Notable differences can be seen compared to the Kanopus results. Indeed, the Kanopus processing showed a much higher proportion of roads and buildings (due to inconsistencies with bare soil) and fewer coastlines. This can be explained by the fact that the Sentinel-2 SOM tended to misclassify the 'coastline' class, confusing it with buildings and roads

Regarding the map generated with Landsat spectral bands, significant discrepancies are observed. Visually, it is immediately apparent that the results are unusable. Almost the entire urban area is misidentified as 'bare soil,' although forests and roads are relatively well represented. Buildings are virtually unrecognized, and water is incorrectly detected across the entire territory. This high level of imprecision is almost certainly due to the coarse spatial resolution of the Landsat spectral bands.

As expected, the land cover distribution reflects these inconsistencies, with 65% of the territory classified as either water or bare soil. The road coverage remains somewhat consistent with previous results. However, the percentages for buildings, forest, and low vegetation are extremely low, as clearly evidenced by the map.

5. CONCLUSION

This remote sensing study applied to the city of **Klaipėda** made it possible to analyze land use using several satellite datasets, as well as to assess their relevance in an urban and peri-urban context. The different classifications carried out using Sentinel-2, Landsat, and Kanopus imagery highlight a territorial structure largely dominated by natural environments, particularly forests, low vegetation, and water bodies, which constitute a relatively stable framework over the studied time period.

The results obtained from Sentinel-2 data, and more specifically through the *Decision Forest* algorithm, appear to be the most consistent from both a visual and statistical perspective. They allow for an overall satisfactory discrimination between urban and natural

classes, despite recurring confusion between buildings, roads, and bare soil, especially in port areas and along the coastline. These limitations are largely related to the spectral similarity of these surfaces, as well as to the choices made during the model training phase.

The comparison between sensors clearly highlights the influence of spatial resolution on classification quality. Kanopus data provide a very high level of spatial detail but tend to accentuate certain classification errors, particularly for artificial surfaces. Conversely, Landsat imagery proves to be too coarse for a fine-scale analysis of Klaipėda's urban fabric, leading to results that are difficult to exploit. In this context, Sentinel-2 appears to be the best compromise between spatial resolution, class stability, and overall consistency of the results.

The projections carried out using a Markov model demonstrate good reliability in the short term, as illustrated by the 2024 forecast, which is very close to the observed reality. The scenarios for the 2035 horizon suggest a moderate evolution of urbanization, characterized by a slight increase in built-up areas and a relative stability of major natural units. However, longer-term projections, particularly for 2055, show a progressive loss of precision with the emergence of spatial inconsistencies, illustrating the inherent limitations of this type of modeling as the temporal horizon extends.

Overall, this work highlights the relevance of remote sensing tools for diachronic land-use analysis and for modeling its evolution. Applied to the city of Klaipėda, this approach provides a global and coherent overview of territorial dynamics, while also emphasizing the need to remain critical of the produced results, particularly when dealing with long-term projections or high spatial resolution classifications.

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